

DealFactor Website Product Specification

1. Product Overview

Working name: DealFactor

Working tagline: Every factor behind every deal.

DealFactor will be a public real-estate discovery and analysis website designed primarily for residential real-estate investors. It will also provide a simpler but complete experience for home buyers.

The website's main purpose is to help users:

1. Find investment properties using natural-language AI search.
2. Determine whether a property is genuinely a good investment.
3. Find and analyze Section 8 investment opportunities.
4. Compare multiple properties, financing structures, and investment strategies.
5. Make property information from multiple approved sources easier to understand.
6. Help home buyers evaluate affordability, condition, neighborhood fit, resale potential, and ownership risks.

DealFactor must prioritize truthful, transparent analysis over overly positive recommendations. It must clearly distinguish verified facts, source data, estimates, assumptions, user inputs, and unavailable information.

2. Launch Positioning

Primary audience

DealFactor will be designed first for:

- Beginner investors searching for their first rental property
- Experienced individual real-estate investors
- Section 8 investors
- Small landlords and residential investors

The interface must be simple enough for beginners while providing the detailed calculations, comparable sales, risks, assumptions, and scenario tools experienced investors expect.

Secondary audience

Home buyers who plan to live in the property they purchase.

The Home Buyer experience will be complete at launch but will be less prominent and less specialized than the investor experience.

Core problems solved

DealFactor's main priorities are:

1. Helping investors accurately determine whether a property is a good deal.
 2. Helping investors find and analyze Section 8 opportunities.
 3. Combining useful property information from multiple sources into a clear interface.
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3. Geographic Coverage

Launch coverage

DealFactor will use a hybrid geographic launch:

- **Nationwide:** Basic property search, address lookup, property facts, photos, standard calculators, and supported baseline analysis.
- **Detroit:** First fully supported advanced investor-analysis and Section 8 market.

Outside Detroit, the website may provide analysis based on the information available, but it must clearly identify limited local data and lower confidence.

Future expansion

The system must support expansion by:

- State
- Region
- Metro area
- County
- City
- ZIP code

Smaller states may be activated statewide. Larger states such as California, Texas, and Florida may be activated in stages.

The website architecture must support future nationwide advanced analysis without rebuilding the core platform.

4. Property Types

DealFactor will support residential properties with one to four units:

- Single-family houses
- Condominiums
- Townhouses
- Duplexes
- Triplexes
- Fourplexes

Not included at launch:

- Properties with five or more units
- Commercial real estate
- Vacant land

These property types may receive separate specialized analysis systems later.

5. Property Data Sources

The production website will connect to authorized APIs for:

- Realtor.com
- Zillow
- Redfin
- HUD and HUD User data
- Local housing-authority data when available
- Geoapify for location, nearby places, routes, and travel times
- Additional reputable public or licensed datasets when needed

Sample property data may be used during development. All data providers must connect through a replaceable property-data layer so live APIs can be added or changed without rebuilding the interface or analysis tools.

API keys must remain on the secure backend and must never be exposed in browser code.

API licensing dependency

Before public launch, every property-data integration must be reviewed to confirm that DealFactor is permitted to:

- Retrieve the data
- Display it publicly

- Store or cache it
 - Combine it with other sources
 - Use it in AI-generated analysis
 - Display listing photos and agent information
 - Show links back to the original listings
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6. Conflicting Property Data Rules

When property sources disagree, DealFactor will apply the following rules.

Majority rule

When two of the three sources report the same value, use the majority value.

Example:

- Realtor.com: 1,500 square feet
- Zillow: 1,500 square feet
- Redfin: 1,620 square feet

Displayed value: **1,500 square feet**

Source-priority rule

When all available sources conflict, use this priority:

1. Realtor.com
2. Zillow
3. Redfin

If Realtor.com is unavailable and Zillow conflicts with Redfin, use Zillow.

Internal recordkeeping

DealFactor must preserve all source values internally even when only one selected value is displayed.

The full source history will be available through **View Data Sources**.

Missing information

Missing information must display:

- Not available
- Insufficient data

- Unverified listing claim
- Low confidence

The AI must never invent missing property facts.

7. Property Refresh Rules

Automatic refresh

- Active listings: twice per day
- Off-market properties: once per week

Every property must display its last-updated date and time.

Manual refresh

Signed-in users receive three manual refreshes per day at launch.

The administrator can change this limit.

The interface must show:

- Remaining refreshes
- Whether a refresh is currently running
- The next scheduled automatic update

A refresh must update the property facts first and then recalculate any affected scores, scenarios, and AI analysis.

API outages

If one provider is unavailable but the necessary information remains available from another source, DealFactor will continue silently.

A user-facing warning is required only when the requested information cannot be retrieved from any relevant source.

Stored recent data may be used when appropriate, but its age must be shown in View Data Sources.

8. Primary Navigation

Desktop navigation

Left side:

- DealFactor logo
- Home
- Investor Search
- Home Buyers
- Deal Analyzer
- My Deals

Right side:

- Help
- About Us
- Contact Us
- Notification bell
- Sign In, Create Account, or Profile

The logo must be clickable and return users to Home.

The top navigation must remain accessible while users scroll.

Mobile navigation

Desktop remains the main development priority.

Mobile will include:

- Logo
- Menu
- Search access
- Profile access
- Simplified navigation
- Optional bottom navigation for Home, Investor Search, Deal Analyzer, and My Deals

Tablet behavior

- Larger tablets: simplified desktop layout
 - Smaller tablets: mobile layout
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9. Landing Page

The landing page must clearly communicate that DealFactor supports both investors and home buyers while prioritizing the investor AI search.

Main investor AI search

The most prominent element will be a large natural-language search box.

Permanent prompt:

What kind of real-estate deal are you looking for?

Rotating examples underneath may include:

- Find Section 8 properties in Detroit with positive cash flow.
- Find beginner-friendly rental properties under \$150,000.
- Find a four-unit property with at least 50% occupancy.
- Find tenant-occupied homes with strong cash-on-cash returns.

Main audience buttons

Directly below the AI search:

- Explore Investor Properties
- Find a Home to Buy

Both buttons must remain prominent, but smaller than the investor AI search.

Analyze a Specific Property

A noticeable but secondary button will say:

Analyze a Specific Property

Clicking it expands an address field directly on the landing page.

The user enters only an address and clicks Search. No additional information is required.

After submission, the user is taken to the Deal Analyzer or general property page.

Supporting landing-page content

The landing page should include:

- Clear explanation of how DealFactor works
 - Investor-analysis benefits
 - Section 8 capabilities
 - Home Buyer tools
 - Data-transparency explanation
 - Trust and disclaimer content
 - Calls to action
 - Contact and About links
-

10. Investor Search

Primary search method

The Investor Search page will lead with natural-language AI search.

Example:

Find tenant-occupied Section 8 properties in Detroit with at least three bedrooms and positive monthly cash flow.

Investment-type selections

Users can select quick strategies such as:

- Section 8
- Long-term rental
- Tenant-occupied
- Multifamily
- Fix-and-flip
- Appreciation
- Lower risk
- Beginner-friendly

Selecting a strategy will:

1. Immediately apply a quick filter.
2. Allow the user to add natural-language details.
3. Provide access to a specialized strategy view when applicable.

Broad-search follow-ups

When a search is broad, the AI may ask one or two optional follow-up questions.

A visible **Skip and Search Now** option must always be provided.

Default filters

- Location
- Price range
- Property type
- Bedrooms
- Bathrooms

More Investor Filters

- Section 8 status
- Occupied or vacant
- Current monthly rent
- Number of units
- Property condition
- Renovation needs
- Minimum monthly cash flow
- Minimum cap rate
- Minimum cash-on-cash return
- Appreciation potential
- Risk level
- Minimum investor score
- Days on market
- Estimated total cash required

Search-result ranking

Ranking priority:

1. Current written search
2. Selected investment goals
3. Saved profile preferences

Manual sorting options include:

- Cash flow
- Cap rate
- Cash-on-cash return
- Risk
- Appreciation potential

- Price
- Overall deal score
- Days on market

Search-result views

Users may switch between:

- Grid
- List
- Map
- Split view

Desktop default: split view

Mobile default: property cards

Investor property cards

Default card information:

- Main photo
- Price
- Address
- Bedrooms
- Bathrooms
- Square footage
- Property type
- Save to My Deals heart

The card must also show two or three short recommendation reasons, such as:

- Strong projected cash flow
- Matches your Section 8 goal
- Below estimated market value

An expandable **Quick Numbers** section includes:

- Monthly cash flow
- Cap rate
- Cash-on-cash return
- Total cash required
- Overall investor score
- Risk level
- Occupancy
- Section 8 status

11. Home Buyers Search

Primary search method

Natural-language AI search will be the main feature.

Example:

Find move-in-ready three-bedroom homes in a safe neighborhood near good schools under \$700,000.

Traditional filters appear underneath.

Broad-search follow-ups

The AI asks follow-up questions only when the request is too broad to create useful results.

Questions must remain brief, optional, and easy to skip.

Default filters

- Location
- Price
- Bedrooms
- Bathrooms
- Property type

More Filters

- Square footage
- Lot size
- Year built
- Move-in readiness
- School information
- Commute
- Neighborhood safety
- HOA fees
- Parking or garage
- Accessibility features
- Property condition
- New construction
- Days on market
- Open houses
- Amenities
- Estimated monthly payment

- Down payment
- Property taxes
- Homeowners insurance

Ranking

The user's current search is the strongest factor.

Profile preferences may influence results but must carry less weight than the current request.

Default sorting: **Best Overall Match**

Additional sorting:

- Price
 - Monthly payment
 - Newest listing
 - Home Buyer score
 - Square footage
 - Days on market
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12. Property Opening Experience

Every property shown anywhere on DealFactor must be clickable.

Quick-view panel

Clicking a property opens a large quick-view panel containing:

- Photos
- Price
- Address
- Status
- Bedrooms
- Bathrooms
- Square footage
- Key financial numbers
- Main score
- Short AI summary
- Save to My Deals
- Compact Ask About This Deal or Ask About This Home action
- View Full Analysis button

Full property page

The complete property page contains:

- Full photo gallery
- Complete property facts
- Listing and price history
- Source links
- Comparable sales
- Financial calculations
- Detailed AI analysis
- Editable assumptions
- Saved scenarios
- Environmental-risk section
- Section 8 tools when applicable
- Property-specific AI assistant
- View Data Sources side panel

Context-based analysis

If opened from Investor Search:

- Show Investor Analysis only.

If opened from Home Buyers:

- Show Home Buyer Analysis only.

If opened from Deal Analyzer, address search, My Deals, shared links, or another general area:

- Show a concise summary of both perspectives.
- Provide access to both full analyses.

For active listings, both analyses may be available in general contexts.

For off-market properties:

- Show all available property information.
 - Show off-market status prominently.
 - Show estimated fair-market value when possible.
 - Do not present a full purchase recommendation as if the property were currently offered for sale.
 - Allow users to manually add deal information if they are considering an off-market transaction.
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13. Investor Property Analysis

Top summary

The full investor page begins with a concise but information-rich summary.

It includes:

- Overall investor score
- Likely score range
- Recommendation
- Asking price
- Estimated fair value
- Price difference
- Monthly and annual cash flow
- Cap rate
- Cash-on-cash return
- Total cash required
- Expected rent
- Risk level
- Appreciation potential
- Leasing outlook
- Resale outlook
- Section 8 status and suitability
- Main strengths
- Main risks
- Missing or low-confidence information

The summary must remain concise. The detailed analysis is the primary specialty of DealFactor.

Detailed sections

Desktop will use expandable sections with side navigation.

Mobile will use expandable sections and a section selector.

Sections include:

1. Financial Performance
2. Pricing and Comparable Sales
3. Risk Analysis
4. Neighborhood and Market
5. Rental and Leasing Outlook
6. Appreciation and Resale
7. Property Condition
8. Renovation Analysis

- 9. Future Strategies
- 10. Section 8
- 11. Environmental Risk
- 12. Data Sources
- 13. Assumptions

Investor category scores

Applicable category scores include:

- Cash flow
- Risk
- Section 8 suitability
- Appreciation
- Flip potential
- Beginner suitability
- Balanced deal quality

All applicable scores must be shown even when the user has chosen one main goal.

14. Investor Goals and Personalization

Users may select one or more goals:

- Highest cash flow
- Lowest risk
- Best Section 8 opportunity
- Best appreciation
- Best flip potential
- Best for a beginner
- Best balanced deal
- Other

The Other option lets the user describe a custom goal.

Default behavior

If goal selection is skipped:

1. Use saved profile preferences.
2. If no preferences exist, use Best Balanced Deal.

Conflicting goals

When selected goals conflict, DealFactor must:

- Explain the tradeoff
- Create a personalized overall recommendation
- Continue showing individual category rankings
- Allow optional advanced weighting

Advanced users may adjust priority weights. Beginners will not be required to do so.

Inapplicable categories

DealFactor must not force a score or ranking when a category is irrelevant or unsupported.

Use:

- Not applicable
 - Insufficient data
 - Low confidence
-

15. Comparable Sales Engine

DealFactor will analyze recently sold comparable properties from approved property sources.

The engine should consider:

- Distance
- Sale date
- Property type
- Bedrooms
- Bathrooms
- Square footage
- Lot size
- Age
- Condition
- Renovation status
- Number of units
- Neighborhood similarity

Comparable-sales output

- Selected comparable properties
- Why each was chosen

- Sale price
- Price per square foot
- Distance
- Sale date
- Similarities and differences
- Current estimated fair value
- Likely value range
- Confidence level

User controls

The AI chooses comps automatically.

Advanced users may:

- Remove comps
- Add other sold properties
- Recalculate the valuation

Flip analysis

When applicable, show:

- Purchase price
- Renovation estimate
- Holding costs
- Selling costs
- Estimated after-repair value
- Projected profit
- Projected return
- Confidence range

16. Ask About This Deal

A prominent property-specific AI assistant will appear only on investor or general property pages.

It must have access to:

- Current property facts
- Comparable sales
- Financial calculations
- Scores
- Assumptions
- Market research

- Section 8 information
- Saved scenarios

It must not mix property-specific information from another property.

It may use the user's general profile preferences.

Placement

- Visible chat box near the top of the full property page
- Floating button while scrolling
- Compact button in quick view
- Full chat opens on the full property page

Example questions

- Why did this property receive this score?
- Recalculate with 10% down and 6.5% interest.
- What rent is needed for \$500 monthly cash flow?
- What are the three biggest risks?
- Would this work better as Section 8?
- Explain cash-on-cash return simply.
- Create a renovation-and-hold scenario.

When the user changes assumptions through chat, DealFactor must update structured calculations and offer to save the result as a named scenario.

Chat history

- Signed-in users: automatically saved by property
- Guests: temporary until the browser tab closes
- Users can delete saved conversations

17. Home Buyer Analysis

Overall Home Buyer evaluation

The analysis must include:

- Overall Home Buyer score
- Affordability score
- Condition and move-in-readiness score
- Neighborhood score
- School score

- Safety score
- Commute score
- Walkability score
- Amenities score
- Resale score
- Ownership-risk score

Monthly-cost analysis

Include:

- Principal
- Interest
- Property taxes
- Homeowners insurance
- HOA fees
- Estimated maintenance
- Total estimated monthly housing cost

All assumptions must be editable.

Upfront cost analysis

Include:

- Down payment
- Closing costs
- Prepaid taxes and insurance
- Inspection
- Appraisal
- Immediate repairs
- Total estimated cash required

Mortgage rate

Use an accurate current estimated rate based on available reliable information.

Consider:

- Loan type
- Term
- Down payment
- Property location
- Owner occupancy
- Optional credit-score range
- Optional lender quote

Credit-score range must be clearly labeled Optional. Leaving it blank must not block the calculation.

Loan types

When applicable, allow comparisons among:

- Conventional
- FHA
- VA
- Other supported loan types later

Affordability tools

Users may:

- Enter income, debt, and down payment
- Enter a preferred maximum monthly payment
- Skip the affordability process using a visible Skip option

Pricing analysis

Show:

- Underpriced, Fairly Priced, or Overpriced
- Estimated fair-value range
- Dollar difference
- Percentage difference
- Comparable sales
- Confidence level

Negotiation analysis

Show:

- Low, Medium, or High negotiation potential
- Reasons
- Suggested offer-price range
- Possible seller-paid closing costs
- Possible repair credits
- General contingency and closing-term ideas
- Questions to discuss with an agent

The website must clearly state that these are estimates and not legal advice or guarantees.

18. Ask About This Home

Home Buyer property pages will have a separate AI assistant called:

Ask About This Home

It focuses on:

- Affordability
- Mortgage scenarios
- Condition
- Move-in readiness
- Neighborhood fit
- Commute
- Comparable homes
- Resale potential
- Ownership risks

Signed-in users may save named Home Buyer scenarios.

Guest scenarios remain temporary until the tab closes.

19. Neighborhood and Proximity Analysis

DealFactor will use Geoapify and MapLibre at launch.

It should evaluate proximity to:

- Grocery stores
- Schools
- Parks
- Restaurants
- Shopping
- Healthcare
- Public transportation
- Major roads
- Employment areas
- Other common daily destinations

Display:

- Walking distance
- Estimated walking time
- Driving distance

- Estimated driving time
- Short, moderate, or long trip classifications

Personalized destinations

Users may optionally enter:

- Workplace
- School
- Family location
- Healthcare provider
- Other frequent destinations

Destinations may be used once or saved privately to the user's profile.

If skipped, DealFactor uses general neighborhood proximity.

Saved personal destinations must not appear in shared reports unless deliberately included.

20. Move-In Readiness and Property Condition

DealFactor may evaluate:

- Listing descriptions
- Listing photos
- Property age
- Inspection reports
- Uploaded documents
- Available condition fields

It must separate:

- Confirmed condition facts
- Visible possible concerns
- AI estimates
- Unknown conditions

The AI must not claim hidden defects based only on photos.

Output may include:

- Move-in-readiness score
- Known repair needs
- Possible concerns
- Estimated repair range

- Confidence level
 - Recommended professional inspections
-

21. Renovation Analysis

Renovation estimates will:

- Default to Standard renovation quality
- Allow users to switch to Budget Rental or High-End
- Show a likely cost range
- Use local labor and material estimates
- Include editable line items
- Include an automatically calculated contingency

Factors include:

- Property size
- Property age
- Visible condition
- Local labor costs
- Material costs
- Permits
- Roof
- HVAC
- Plumbing
- Electrical
- Foundation
- Kitchen
- Bathrooms
- Flooring
- Paint
- Windows
- Appliances
- Exterior
- Landscaping
- Cleanup

Contingency reserve

The default contingency varies based on:

- Property age
- Condition
- Scope
- Inspection data

- Data confidence

Users may edit the contingency.

All connected returns and scores must recalculate after changes.

22. Deal Analyzer

Deal Analyzer will be a permanent top-navigation page.

It will support:

- One-property analysis
- Multi-property comparison
- Properties found through DealFactor
- Properties searched by address
- Off-market properties
- Manually entered outside deals
- Uploaded listing or investment documents

One-property mode

The user receives:

- Property facts table
- Financial analysis
- Investor and Home Buyer perspectives when appropriate
- Scores
- Risks
- Comparable sales
- Follow-up AI assistant
- Editable assumptions
- Saved scenarios

Multi-property mode

Users can compare multiple properties in a structured workspace.

Automatic AI comparison brief

The page immediately explains:

- Strongest overall option
- Best cash flow
- Lowest risk

- Best Section 8 option
- Best appreciation potential
- Best beginner option
- Main weakness of each deal
- Missing or uncertain information

Goal controls

- Best Overall
- Highest Cash Flow
- Lowest Risk
- Best Section 8 Deal
- Best Appreciation
- Best for Beginners
- Custom goal

Comparison table

Potential metrics:

- Purchase price
- Fair value
- Total cash required
- Down payment
- Mortgage payment
- Expected rent
- Expenses
- Cash flow
- Cap rate
- Cash-on-cash return
- Appreciation potential
- Risk
- Condition
- Renovation needs
- Leasing demand
- Resale demand
- Section 8 suitability
- Data confidence
- Overall score

Rankings

Each applicable category receives a separate ranking.

Example:

- Highest cash flow: B, A, C
- Lowest risk: A, C, B

- Best appreciation: C, B, A
- Best balanced deal: A, B, C

Comparison AI assistant

Users may ask:

- Which is safest with \$30,000 available?
 - Recalculate all with 10% down.
 - Which is easiest to manage remotely?
 - Which has the best risk-adjusted return?
 - What should I verify before making an offer?
-

23. Editable Assumptions and Scenarios

Analyses load immediately with automatic assumptions.

A clear **Edit Assumptions** button allows users to modify:

- Down payment
- Interest rate
- Loan term
- Expected rent
- Vacancy rate
- Management fee
- Maintenance reserve
- Insurance
- Property taxes
- HOA fees
- Utilities
- Renovation costs
- Other relevant expenses

Update process

1. Preview updated results.
2. Apply temporarily.
3. Cancel.
4. Save as a named scenario.

Examples:

- 10% Down
- HUD Rent
- Self-Managed

- Renovate and Hold
- Flip Scenario

Users may compare saved scenarios side by side.

24. Section 8 Status and Entry Points

Confirmed or reported involvement

Possible status labels:

- Section 8 Mentioned — Unverified
- Section 8 Tenant Reported
- Active Section 8 Tenant — Confirmed
- HAP Contract Confirmed

Most listing-description claims will be labeled unverified unless additional evidence exists.

Action button:

View Section 8 Analysis

No current Section 8 involvement

Eligible residential investment properties display:

Analyze as Section 8

This opens a clearly labeled hypothetical analysis.

It must not imply:

- Guaranteed housing-authority approval
- Guaranteed inspection approval
- Guaranteed voucher tenant
- Guaranteed HUD rent
- Guaranteed HAP contract

The Section 8 entry point appears in:

- Investor Analysis
- General property pages
- Deal Analyzer

It does not appear inside the Home Buyer-only experience.

25. Section 8 Analysis

Opening experience

Clicking the Section 8 action opens a concise summary panel.

The panel shows:

- Status
- Rent scenarios
- Cash flow
- Cap rate
- Cash-on-cash return
- Balanced Section 8 score
- Personalized Section 8 fit score
- Main risks
- Main opportunities

A prominent **View Full Section 8 Analysis** button opens the complete analysis within the property page.

Rent scenarios

Display side by side:

1. Current confirmed rent, when available
2. HUD benchmark rent
3. Estimated open-market rent

Also display separately:

- Local housing-authority payment standard, when available
- Estimated rent-reasonableness range

Every amount must be labeled as:

- Confirmed
- Estimated
- Benchmark
- Unverified
- Not available

Missing current rent

When a listing reports a Section 8 tenant but does not provide rent:

- Current rent: Not available
- Calculate a HUD benchmark scenario
- Calculate a market-rent scenario
- Do not present either as confirmed current rent

Tenant and housing-authority rent portions

When available, show:

- Tenant-paid portion
- Housing-authority-paid portion
- Total monthly rent

Missing portions display Not available.

Utility allowance

Include utility allowance calculations only when the allowance is stated or confirmed.

When available, show:

- Before allowance
- After allowance
- Monthly difference
- Annual difference

Do not guess an allowance.

Rent increase analysis

Calculate:

- Monthly increase
- Annual increase
- Percentage increase
- Updated cash flow
- Updated cap rate
- Updated cash-on-cash return
- Updated score

Clearly state that the final approved rent depends on housing-authority review.

26. Section 8 Financial Calculations

For each Section 8 analysis, support:

- Address
- Occupancy status
- Square footage
- Bedrooms
- Bathrooms
- Total acquisition cost
- Down payment
- Loan amount
- Interest rate
- Loan term
- Monthly mortgage payment
- Monthly rent
- Annual rent
- Monthly property taxes
- Monthly insurance
- Monthly property management
- Monthly maintenance reserve
- Monthly HOA
- Monthly vacancy reserve
- Monthly utilities
- Total monthly operating expenses
- Total annual operating expenses
- Monthly NOI
- Annual NOI
- Monthly cash flow
- Annual cash flow
- Price per square foot
- Cap rate
- Cash-on-cash return

Formulas

Loan Amount

Total Acquisition Cost – Down Payment

Monthly Mortgage Payment

$PMT(\text{Interest Rate} \div 12, \text{Loan Term} \times 12, -\text{Loan Amount})$

Annual Rental Income

Monthly Rental Income $\times$ 12

Total Monthly Operating Expenses

Property Taxes + Insurance + Property Management + Maintenance Reserve + HOA + Vacancy Reserve + Utilities

Total Annual Operating Expenses

Total Monthly Operating Expenses $\times$ 12

Monthly NOI

Monthly Rental Income – Total Monthly Operating Expenses

Annual NOI

Annual Rental Income – Total Annual Operating Expenses

Monthly Cash Flow

Monthly NOI – Monthly Mortgage Payment

Annual Cash Flow

Monthly Cash Flow $\times$ 12

Price per Square Foot

Total Acquisition Cost $\div$ Square Footage

Cap Rate

Annual NOI $\div$ Total Acquisition Cost

Cash-on-Cash Return

Annual Cash Flow $\div$ Down Payment Amount

Important calculations must be performed by tested application code, not improvised by the AI.

27. Section 8 Specialized Risk Analysis

Include:

- Approval-risk rating
- Property-condition concerns
- Likely inspection issues when evidence exists
- Missing documentation
- Local housing-authority considerations
- Repair requirements supported by available data
- Tenant demand
- Leasing outlook
- Payment-interruption risk
- HAP abatement risk

- Resale potential
- Appreciation potential
- Overall recommendation

Inspection guidance

Provide a general educational inspection checklist.

Only identify property-specific possible repairs when supported by:

- Photos
- Listing description
- Inspection report
- Uploaded document
- Other reliable data

When condition information is unavailable, do not guess. Show a brief notice explaining that repairs may be required and an official inspection is necessary.

Approval and Lease-Up Process

Provide a collapsed dropdown titled:

Approval and Lease-Up Process

It may include:

- Finding a voucher tenant
- Tenancy approval request
- Rent-reasonableness review
- Inspection
- Repairs
- Reinspection
- Lease signing
- HAP contract
- Payment commencement

Use local estimated timing when reliable data exists. Otherwise, use clearly labeled general ranges.

Payment interruption and HAP abatement

Include:

- Possible causes
- Inspection-failure risks
- Repair requirements

- Approximate deadlines when known
- Estimated cash-flow impact
- Confidence level

Do not state that abatement will occur without evidence.

Tenant and payment history

Show this section only when reliable information is available.

Possible fields:

- Length of tenancy
- Rent history
- Renewal history
- Prior vacancies
- Reported inspection issues
- Reported HAP issues

No tenant names, contact information, or identifying information may be displayed.

28. Section 8 Deal Scores

Show:

- Balanced Section 8 Deal Score
- Personalized Section 8 Fit Score

Category scores include:

- Financial performance
- Approval risk
- Tenant demand
- Property condition
- Rent potential
- Payment-interruption risk
- Resale and long-term potential

All scores must include:

- Reasoning
- Assumptions
- Confidence
- Likely range when confidence is low

Section 8 comparisons

Comparable-deal priority:

1. Same ZIP code
2. Same city
3. Same metro area
4. Similar nearby market

Show the distance and similarity of every comparable deal.

29. Low-Confidence Analysis

When important information has low confidence, display:

Estimated score: 72/100

Likely range: 68-76

Low-confidence data

The low-confidence warning is clickable.

It explains:

- Which inputs are uncertain
- Why they are uncertain
- Which sources were used
- How the uncertainty affects the analysis
- What the user should verify independently

DealFactor will generally allow the analysis to proceed using conservative assumptions rather than blocking it.

The administrator may change this behavior per market.

30. Accounts and Profiles

Sign-in options

- Email and password
- Google sign-in

Email verification is not required.

Users may begin using their accounts immediately.

Profile preferences

Profile setup is optional and may be skipped.

Possible preferences:

- Experience level
- Investment strategy
- Section 8 interest
- Preferred markets
- Property types
- Risk tolerance
- Cash flow versus appreciation preference
- Available down-payment range
- Preferred monthly cash flow
- Self-management versus property management

If no preferences exist, DealFactor defaults to Best Balanced Deal.

Financial profile information

Users may optionally save ranges for:

- Income
- Debt
- Available down payment
- Credit score
- Preferred monthly payment
- Loan type
- Loan term

Exact figures may be entered temporarily in calculators.

Exact figures should not be saved unless the user deliberately chooses to save them.

31. Guest Usage

Guests may:

- Search properties
- Use Deal Analyzer
- View analysis
- Change assumptions temporarily

- Use property-specific AI chats
- Submit issue reports

Guest analysis remains available only while the browser tab is open.

Refreshing may preserve the active session, but closing the tab removes it.

The page should warn users before leaving with unsaved work.

When a guest clicks **Save to My Deals**, DealFactor prompts them to sign in or create an account.

After sign-in, the active property and analysis transfer into My Deals without restarting.

32. My Deals

Users may save a property using a heart button labeled:

Save to My Deals

The button appears on:

- Search-result cards
- Quick-view panels
- Full property pages
- Deal Analyzer
- Comparison results
- Address-based analyses

Saved-property information

Each saved property may show:

- Photo
- Address
- Price
- Status
- Overall score
- Cash flow
- Cap rate
- Cash-on-cash return
- Section 8 suitability
- Saved scenarios
- Date saved
- Last update
- Price or status changes

Folders

Users may create folders such as:

- Detroit Shortlist
- Section 8 Deals
- Properties to Tour
- Fix-and-Flip Options

Saving is immediate. Folder selection is optional.

Progress stages

- Saved
- Researching
- Contacted
- Under Contract
- Purchased
- Rejected

Notes and tasks

Users may add private notes.

Each deal may have an optional collapsible checklist opened through a button such as:

- Open Tasks
- View Checklist

Users can add, edit, complete, and delete tasks.

33. Notifications

Access

- Notification bell in top navigation
- Full notification history inside My Deals

Default notifications

Enabled by default:

- Price changes
- Listing-status changes

- Major deal-score changes

Users may configure notifications:

- Globally
- Per saved property

Notifications are in-app only at launch.

Email and text notifications are future features.

34. Manual Deal Entry and Uploads

When a property cannot be retrieved, users may enter it manually.

Simple starting form

- Address
- Purchase price
- Rent
- Bedrooms
- Bathrooms
- Square footage

Add More Details

Optional advanced fields include:

- Financing
- Expenses
- Occupancy
- Property condition
- Renovation costs
- Section 8 information
- Number of units
- Taxes
- Insurance
- HOA
- Utilities

Upload-supported formats

- PDF
- Images and screenshots
- Spreadsheets

The AI may extract and prefill deal information.

Users must review and confirm extracted values before analysis.

Uncertain fields display:

Needs review

Document retention choice

After extraction, ask:

Save this document to the deal?

Options:

- Save to deal
- Delete document

Deleting the file may preserve confirmed extracted data.

Sensitive information

Before upload, warn users to remove unnecessary personal information.

DealFactor should attempt to detect and redact:

- Tenant names
- Phone numbers
- Email addresses
- Social Security numbers
- Bank information
- Signatures

Automatic redaction must be presented as assistance, not a guarantee.

35. Sharing and Reports

Signed-in users may:

- Generate a private view-only link
- Set an optional password
- Set an expiration date
- Disable the link

- Download a property-analysis report

Shared viewers cannot edit the analysis.

Private notes and tasks are excluded by default.

Uploaded documents are excluded unless explicitly included.

Social preview

Private property-analysis links use a generic DealFactor preview.

The preview must not expose:

- Address
- Property photo
- Scores
- Financial results
- User identity

Possible preview text:

A private property analysis has been shared with you.

36. General Help AI

The permanent Help AI assistant will:

- Explain how to use DealFactor
- Explain calculations and terminology
- Assist with accounts and saved features
- Guide users to the correct page
- Help users begin a search

It will not provide property-specific analysis.

Property-specific analysis belongs to:

- Ask About This Deal
- Ask About This Home
- Deal Analyzer

Placement

- Help link in top navigation

- Small unobtrusive chat button in the bottom corner of every page
-

37. Contact Us

Keep a dedicated Contact Us page.

Use one simple contact form with a request-type dropdown:

- General question
- Technical support
- Incorrect property data
- AI analysis issue
- Billing or account help
- Partnership or business inquiry
- Press or media
- Other

Public email is shown only for:

- Business
- Partnerships
- Media

Technical and account support use the contact form.

The Help AI may transfer unresolved issues into the contact form and prefill a summary.

38. Report an Issue

Every analysis includes a simple **Report an Issue** button.

The form asks for a brief explanation.

DealFactor automatically attaches:

- Property
- Analysis version
- Assumptions
- Data sources
- Date
- Relevant system information

Guests may report issues.

In the admin dashboard, they appear as:

Guest User

Guests may provide an optional email address.

Signed-in users may receive in-app report updates.

39. About Us

The About Us page will initially present DealFactor as a company and product.

Founder name, photo, and story will be added later.

The page should explain:

- Mission
- Why DealFactor was created
- Investor and Home Buyer value
- Section 8 focus
- Data approach
- AI analysis approach
- Source transparency
- Confidence levels
- Missing-data policies
- AI limitations
- Commitment to honest analysis

The language should be approachable rather than overly corporate.

40. Administrator Dashboard

The dashboard must support future staff roles, though only the owner will have access initially.

Prepared roles:

- Full administrator
- Market-data manager
- Support
- Content editor

Administrator features

- User management
- Issue reports
- Support requests
- Data-source status
- API configuration
- AI monitoring
- Market coverage
- Market expansion
- Content management
- Legal and disclaimer content
- Usage analytics
- Notification settings
- Refresh limits
- Backup and restore
- Activity logs

Market coverage statuses

- Not configured
- In development
- Internal testing
- Limited beta
- Fully supported
- Temporarily paused

Market-expansion workflow

1. Enter state, region, county, city, or ZIP code.
2. Retrieve connected property data.
3. Research local assumptions.
4. Check source coverage.
5. Generate suggested assumptions.
6. Generate sample analyses.
7. Review confidence and warnings.
8. Approve, edit, request more research, limit, or reject.
9. Publish the market.

The administrator chooses the number of sample analyses each time. Default: 10.

Automated market research

Research items may include:

- Vacancy
- Insurance

- Management fees
- Maintenance reserves
- Appreciation history
- Rental demand
- Leasing speed
- Renovation costs
- Neighborhood risk
- Section 8 requirements

For each suggested assumption, show:

- Suggested value
- Supporting range
- Sources
- Source reliability
- Data date
- Geographic relevance
- Confidence
- Approve
- Edit
- Request more research

The administrator approves final assumptions, not every individual source.

41. Content Management

The admin dashboard will include a content-management system.

Editable without code:

- Landing-page text
- About Us
- FAQs
- Help articles
- Disclaimers
- Contact information
- Search examples
- Announcements
- Supported-market notices
- Reusable labels

Features:

- Preview before publishing
- Version history
- Restore earlier content

Financial formulas, scoring logic, API settings, and security settings must remain separated from ordinary content editing.

42. Administrator Security

Admin login requires:

1. Email and password
2. SMS verification code to the primary phone

Recovery methods:

- Verified backup phone number
- Verified recovery email

Changing recovery methods requires current authentication.

Sensitive admin actions require recent reauthentication.

Examples:

- Backup restore
- Market publication
- Security changes
- Data-source changes

Admin activity log

Maintain a complete searchable log containing:

- Summary
- Date and time
- Administrator
- Affected user, property, market, or setting
- Previous value
- Updated value
- Security and recovery events

Verification codes must never be stored in the activity log.

43. Data Sources Panel

Property pages will include a small **View Data Sources** button near the property facts.

Clicking it opens a side panel showing:

- Source for each important fact
- Conflicting values
- Selected value
- Why it was selected
- Last update time
- Confirmed, estimated, description-extracted, or unavailable status

Source labels should not clutter the main property interface.

Original listing links

Primary listing link:

1. Realtor.com
2. Zillow if Realtor.com is unavailable
3. Redfin if both are unavailable

An expandable **View Other Sources** section shows Zillow and Redfin links when available.

44. Environmental Risk

Include a supporting Environmental Risk section.

Potential topics:

- Flood
- Wildfire
- Earthquake
- Severe weather
- Other location-specific hazards

Show:

- Available risk rating
- Explanation
- Source
- Confidence
- Possible insurance impact
- Possible ownership or investment impact
- Recommended professional verification

Environmental risk should influence the overall risk score when financially significant, but it must not become the main focus of the analysis.

45. AI Architecture

Development versus production AI

Claude Code will be used to build and maintain the website.

The live website will use the Anthropic Claude API.

One Anthropic API account may power:

- Investor analysis
- Home Buyer analysis
- Section 8 analysis
- Ask About This Deal
- Ask About This Home
- Deal Analyzer
- Help AI
- Document extraction
- Admin research summaries

Each assistant must have separate instructions, permissions, context, and tools.

Calculation rule

The application code calculates critical numbers.

Claude interprets, explains, compares, summarizes, and answers questions.

The AI must not freely generate core financial calculations when tested code can calculate them.

AI provider flexibility

The AI layer must be modular so another provider may be added or substituted later.

46. Recommended Technical Stack

Frontend

- Next.js
- TypeScript
- Responsive desktop-first interface
- GitHub repository from the beginning

Backend

Supabase will be the core backend at launch:

- PostgreSQL database
- Authentication
- User permissions
- File storage
- Edge Functions
- Queues
- Scheduled jobs
- Realtime progress updates

No separate background-job platform is required initially.

The code should allow one to be added later if workloads become too large.

Hosting

- Vercel or another compatible professional hosting platform

AI

- Anthropic Claude API

Mapping

- Geoapify
- MapLibre

Development tool

- Claude Code

47. Database Persistence and Backups

User information must persist between sessions.

Supabase should store:

- Accounts
- Profiles
- Preferences
- My Deals

- Folders
- Stages
- Notes
- Tasks
- Scenarios
- Chats
- Notifications
- Documents
- Admin settings

Backups

- Weekly restorable backups required
- Use more frequent automatic backups when included by the provider
- Backup uploaded files separately when database backups do not include file contents

Restore workflow

Before restoration:

1. Preview affected records.
2. Show newer information at risk.
3. Prefer record-level recovery.
4. Create a new safety backup.
5. Require administrator approval.

48. Privacy and Account Deletion

Users can delete their accounts.

Deletion removes personal information from active systems immediately, including:

- Profile
- Financial preferences
- My Deals
- Scenarios
- Notes
- Tasks
- Documents
- Saved chats
- Private sharing links

Encrypted backup remnants may remain inaccessible until the normal backup cycle expires.

Download My Data

Before deletion, offer an export containing:

- Profile
- Preferences
- Saved properties
- Folders
- Scenarios
- Comparisons
- Notes
- Tasks
- Chats
- Documents
- Reports
- Shared links

Use common formats such as PDF, CSV, and original upload formats.

49. Legal Requirements

Launch legal pages:

- Privacy Policy
- Terms of Use
- Cookie Policy
- Investment and financial disclaimer
- AI disclaimer
- Property-data disclaimer
- Fair Housing notice
- Accessibility statement

Users accept the main terms and disclaimers:

- When creating an account
- When first using advanced analysis

Do not interrupt users with a long disclaimer before every analysis.

Short contextual notices appear beside:

- Financial estimates
- Mortgage calculations
- AI recommendations
- Neighborhood scores

- Section 8 projections
- Renovation estimates

DealFactor does not provide guaranteed:

- Investment returns
- Property values
- Mortgage terms
- Appraisals
- Inspections
- Legal advice
- Tax advice
- Housing-authority approval
- Seller acceptance

There is no age-verification requirement.

50. Cookies and Analytics

Browser storage

Use only essential cookies or browser storage for:

- Authentication
- Security
- Appearance preference
- Temporary guest analysis
- Abuse prevention

No advertising or cross-site tracking cookies at launch.

Analytics

Use privacy-focused, first-party product analytics.

Track:

- Visitors
- Account creation
- Feature usage
- Search completion
- Saved deals
- Deal Analyzer use
- Property-page engagement
- Errors and drop-off points

Do not record:

- Private search wording
 - Uploaded document contents
 - Personal financial information
 - Private AI conversations
 - Passwords
 - Sensitive account information
-

51. Visual Design

Brand style

DealFactor should combine:

- Professional financial-platform design
- Modern AI styling
- Polished real-estate photography

The design must remain functional and beginner-friendly.

Color palette

- Dark navy
- White
- Subtle gold accents

Gold should be used sparingly for:

- Selected controls
- Important scores
- Key actions
- Premium visual details

Typography

- Elegant headings
- Clean modern body text
- Highly readable financial tables and numbers

Visual content

Use a balanced mix of:

- Property photography
- Maps
- Charts
- Tables
- Subtle AI graphics

Visuals must support the product rather than overwhelm it.

Light and dark modes

Default to the user's device preference.

Allow:

- Light
- Dark
- System

Manual selections override the device preference.

52. Animation and Loading

Animation

Use moderate, polished animations for:

- Page transitions
- Property cards
- Charts
- AI processing
- Expanding panels

Animations must remain fast and functional.

Respect reduced-motion settings.

Loading

Short actions:

- Spinner or subtle indicator

Long actions:

- Step-by-step progress panel

Possible stages:

- Finding properties
- Combining listing data
- Reviewing comparable sales
- Researching market conditions
- Calculating returns
- Evaluating risks
- Generating analysis

Do not show fake precision or inaccurate percentages.

Allow cancellation when practical.

Accuracy and completeness take priority over speed.

53. Accessibility and Responsive Priorities

Desktop is the clear priority.

The desktop version will contain the richest:

- Comparison tables
- Split maps
- Dashboards
- Side navigation
- Advanced filters

Mobile must still support:

- Search
- Property viewing
- Analysis
- Saving deals
- AI assistants

- Account access

Meet standard accessibility requirements without cluttering the interface.

Include:

- Readable contrast
- Keyboard navigation
- Screen-reader labels
- Accessible forms
- Clear errors
- Descriptive image text
- Reduced-motion support

54. Search-Engine and Sharing Settings

At launch:

- DealFactor will be publicly accessible through its URL.
- Search engines will be instructed not to index the website.
- Indexing can be enabled later.

Private pages must always remain excluded:

- My Deals
- Profiles
- Uploads
- Saved scenarios
- AI conversations
- Private reports

Public links should have polished DealFactor social previews.

Private analysis links use the generic non-property preview described earlier.

55. Monetization

DealFactor will be completely free at launch.

No:

- Subscription
- Paywall

- Payment processing
- Upgrade pressure

The architecture should allow monetization later through:

- Paid plans
 - Premium reports
 - Higher limits
 - Referrals
 - Partnerships
 - Other approved methods
-

56. Launch Scope

Fully included launch capabilities

- Nationwide basic property search
- Detroit advanced investor analysis
- Investor AI search
- Home Buyer search and analysis
- Address search
- Deal Analyzer
- One-property and multi-property analysis
- Comparable sales
- Editable assumptions
- Saved scenarios
- Full Section 8 analysis
- Ask About This Deal
- Ask About This Home
- Accounts
- My Deals
- Folders
- Stages
- Notes
- Tasks
- Maps and proximity
- Data-source transparency
- Confidence warnings
- Issue reporting
- Manual deal entry
- Document uploads and extraction
- Private sharing
- Standard downloadable report
- In-app notifications
- Core admin dashboard

- Market-expansion controls
- Content-management system

Simplified at launch

- Admin market-research automation
- Document extraction for uncommon documents
- Analytics dashboards
- Mobile optimization
- Report customization
- Profile weighting controls
- Renovation models outside fully supported markets
- Advanced staff-role interfaces

These foundations should still be designed for later expansion.

57. Future Features

Officially saved for later:

- Paid plans and monetization
 - Email and text property alerts
 - Collaborative deal workspaces
 - Direct property-tour requests
 - Agent partnerships
 - Lender partnerships
 - Property-manager partnerships
 - Fully activated staff roles
 - More advanced mobile optimization
 - Advanced investor analysis outside Detroit
 - Specialized five-plus-unit multifamily analysis
 - Commercial property analysis
 - Vacant-land analysis
 - Additional report formats
 - Deeper analytics
 - More advanced continuous market research
 - Apple sign-in
 - Email verification if later needed
-

58. Testing

Launch testing will include:

- Manual testing
- Automated tests for financial calculations
- Automated tests for major workflows

Required automated coverage:

- Mortgage calculations
- NOI
- Cash flow
- Cap rate
- Cash-on-cash return
- Section 8 scenarios
- Source-conflict rules
- Assumption updates
- Saved scenarios
- Sign-up
- My Deals
- Uploads
- Sharing

The owner will conduct most broader real-world testing.

More extensive security, API-failure, device, and high-traffic testing may be added later.

59. Performance and Traffic

Expected launch traffic:

- Fewer than 10 users

Large-scale infrastructure is not a launch priority.

The architecture should remain organized enough to expand later without requiring an immediate enterprise setup.

Basic pages and stored data should load quickly.

Full analysis may take longer when necessary to retrieve and verify all available data.

DealFactor must never skip available information simply to generate a faster result.

60. Operating Budget

Target initial operating cost:

- Approximately \$50–\$100 per month

Accuracy and reliability are more important than selecting the cheapest service.

Before implementation, prepare a cost breakdown covering:

- Hosting
- Supabase
- Claude API
- Mapping
- Storage
- Property-data APIs
- Supporting research services

Property-data API licensing is the cost most likely to exceed this budget.

If the projected cost exceeds the target, explain:

- Which service causes the increase
- Why it is needed
- Whether a cheaper alternative exists
- What functionality would change under the cheaper option

61. Success Criteria

The first version is successful when:

1. Property data is retrieved and combined accurately.
2. Financial calculations are reliable.
3. The AI clearly explains assumptions and uncertainty.
4. Missing information is not invented.
5. Search results match user requests.
6. Users can identify useful deals worth investigating further.
7. Section 8 analysis provides practical, understandable information.
8. Beginner investors can understand the analysis.
9. Experienced investors can inspect and edit the underlying assumptions.
10. The website works reliably for its initial users.

Large traffic, subscriptions, and revenue are not required to declare the first version successful.

62. Remaining External Dependencies

The following items must be completed outside this specification:

- Final DealFactor logo
- Final domain purchase
- Final founder information
- Zillow API access
- Realtor.com API access
- Redfin API access
- HUD API access
- Verification of data-display and storage rights
- Anthropic API key
- Supabase project
- Geoapify key
- SMS provider for administrator two-factor authentication
- Legal review of policies and disclaimers
- Sample property data for development
- Final scoring formulas and category weights
- Final Detroit market assumptions
- Final production cost review

These dependencies should not require major changes to the product architecture described above.